ENGO 531/632

Advanced Photogrammetric and Ranging Techniques

Laboratory Assignment 5

Terrestrial Laser Scanning and Point Cloud Colourization

Date given: 6 November 2026

Date due: 26 November 2026 at 23.59
Late group submissions will not be accepted;
Late individual submissions lose 10% per day

Marks: This assignment constitutes 15% of your final grade.

Objective

The objective of this lab is to perform a terrestrial laser scanner survey of your chosen object and to colourize the point cloud using imagery captured with your phone camera.

For students in ENGO 531

This assignment is to be performed by the group formed for labs 1-4.

For students in ENGO 632

This is an individual assignment.

Assignment

Laser scanner data collection—general

Capture a point cloud of your object with a single scan using the Department's Trimble X9 laser scanner. The TA and/or the technician will provide instructions about its operation.

Point cloud preparation

Import your dataset into CloudCompare, which is available on all Microlab PCs. CloudCompare is a free software for manipulating point clouds. See https://www.danielgm.net/cc/ for background. A brief demonstration will be provided in the lab, but a very good tutorial (about 11 minutes long) can be found here: https://youtu.be/MQiD4HjhpAU. You can find the wiki for CloudCompare here: https://www.cloudcompare.org/doc/wiki/index.php?title=Main_Page.

Once imported, you should inspect your dataset and crop out unwanted points (there will be many), leaving only points on your object. To do so, use the segmentation tool (scissors icon) in CloudCompare. Export the edited point cloud as a text file. The TA will provide instruction on how to perform the segmentation.

Each line in the output text file corresponds to a point and comprises six fields: X Y Z R G B. Edit your text file to remove the R G B fields, retaining only the X Y Z fields. You can either do this manually (not recommended), using a macro function programmed in a text editor such as TextPad, or by writing a program in MATLAB or other language to do so. *Note by I.D.: or use the Alt key while highlighting unwanted columns of text in Notepad++ and then delete.*

Image orientation

The point cloud you capture is referenced to the laser scanner's internal coordinate system, which does not coincide with the coordinate system defined for your network. To map the camera texture to the point cloud, you need to determine the transformation between the point cloud coordinate system and image space for each the camera image. There are several ways to do this. One is described below.

Identify a set of distinct feature points in the image network. These points should also be readily identifiable in the point cloud. Each point should appear in more than one image. The points should be in a strong geometric configuration. Observe the image coordinates of the feature points in each image in which they appear.

Observe the 3D coordinates of the same points in the point cloud using the Point Picking tool in CloudCompare. These points will be treated as control points that are to be included in a new bundle adjustment of the network. For this new adjustment, do not apply the parameter constraints to the EOPs as you did in lab 4. You can also remove the distance observation. These actions change both the definition of object space and the datum definition. Therefore, you may need to modify the approximate EOPs and tie point coordinates. Make sure the units of the new 3D coordinates match the object space units of your original network.

Backprojection

The new bundle adjustment determines the orientation of your photogrammetric network in the laser scanner coordinate system. Now you can colourize the point cloud by backprojecting each point in the point cloud (this is why careful editing is needed) into the imagery. To do so, use the newly determined EOPs, known IOPs and known TLS object point coordinates to compute x and y image space coordinates for each point in the point cloud using the collinearity equations. Be sure to account for the left-handed coordinate system of pixel space. Create a program in MATLAB or other language to perform the backprojection.

Round the backprojected image point coordinates to obtain integer values for the row (y) and column (x) location. Note that this is essentially nearest neighbour resampling. Assign the R G B values from the image at this location to the X Y Z coordinates and write them to a new text file with the same format as the file produced by CloudCompare.

Note that any given object point may appear in multiple images, which raises the question of which image should be used to obtain the R G B values. There are ways to determine the "best" image to use. Decide upon a method among the members of your group (ENGO 531 only). It need not be the best way—it likely will not be, but this is not important. What is important is to report the method, the reason you chose it and the resulting problems. Whatever method you choose, you must check that the backprojected image point lies within the image boundaries—this is essential. If a point does not lie within the boundaries of any of your images, skip the point.

Submissions

You are to report on the findings of your project, in particular the following:

- Report on the TLS data capture
- Comments on the geometric design of your revised network in terms of the control points Address ZOD, FOD and SOD
- A brief discussion of the model fit
- Discussion of your chosen method to choose which image to use to obtain the R G B values.
- Import the colourized point cloud into CloudCompare and inspect it. Comment on the quality of the colourization
- Explain possible causes for any errors you observe in the colourized point cloud and how the final product could be improved
- List three safety measures that should be followed when operating the laser scanner

Note that the quantities you analyze are for you to decide, but your analyses are expected to be detailed and justified. It is strongly suggested that you include plots and tables to support your observations. Submissions should include field notes and a sketch of the data capture.

Submissions are to be uploaded as a single PDF file to the D2L Dropbox by the specified date and time. For ENGO 531 students, only one group member needs to make a submission but be sure to indicate all members of the group on the title page.

END OF LAB 5